

Finding $\vec{v}_p$:

$\vec{v}_p = \vec{v}_B + \vec{\omega}_{BP} \times \vec{r}_{P/B}$

$v_p\vec{j} = \vec{v}_B + (-\omega_{BP}\vec{k} \times (-L\sin(\phi)\vec{i} + L\cos(\phi)\vec{j}))$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\omega_{BP} \\ -L\sin(\phi) & L\cos(\phi) & 0 \end{vmatrix} = \vec{i}(-\omega_{BP}\cos\phi) + \omega_{BP}\sin\phi$$

$v_p\vec{j} = \vec{v}_B + L\omega_{BP}(\cos(\phi)\vec{i} + \sin(\phi)\vec{j})$

Need $\vec{v}_B$ first:

$\vec{v}_B = \vec{v}_A + \vec{\omega}_{AB} \times \vec{r}_{B/A}$

$\vec{v}_B = -\omega_{AB}\vec{k} \times (r\sin(\theta)\vec{i} + r\cos(\theta)\vec{j})$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\omega_{AB} \\ r\sin\theta & r\cos\theta & 0 \end{vmatrix}$$

$-\omega_{AB} r\cos\theta\,\vec{i} - \omega_{AB} r\sin\theta\,\vec{j}$

Combining:

$v_p\vec{j} = -\omega_{AB} r\cos\theta\,\vec{i} - \omega_{AB} r\sin\theta\,\vec{j} + L\omega_{BP}\cos(\phi)\vec{i} + L\omega_{BP}\sin(\phi)\vec{j}$

split into $\vec{i}$ and $\vec{j}$

$\vec{j}: v_p = -\omega_{AB} r\sin\theta + L\omega_{BP}\sin\phi$

$\vec{i}: -\omega_{AB} r\cos\theta = L\omega_{BP}\cos\phi$

$$\omega_{BP} = \frac{-\omega_{AB} r\cos\theta}{L\cos\phi}$$

$$v_p = \omega_{AB} r\sin\theta + L\sin\phi\left(\frac{-\omega_{AB} r\cos\theta}{L\cos\phi}\right)$$

$$\boxed{v_p = \omega_{AB} r(\sin\theta + \cos\theta\tan\phi)}$$

because $r\sin\theta = L\sin(\phi)$:

$$\boxed{\phi = \sin^{-1}\left(\frac{r}{L}\sin(\theta)\right)}$$